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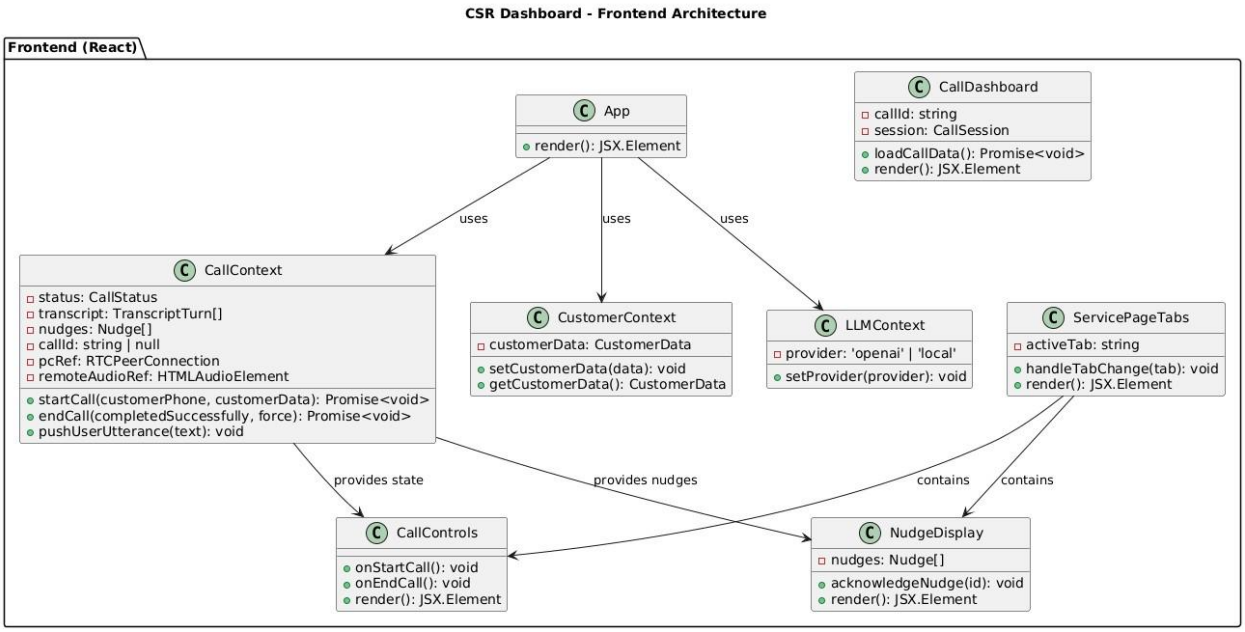
Frontend

React frontend architecture and component relationships. The App component serves as the root and integrates three main context providers: CallContext (manages call state, WebRTC connections, and audio handling), CustomerContext (stores customer information), and LLMContext (manages AI provider selection).

The ServicePageTabs component contains the main call interface, housing CallControls (start/end call buttons) and NudgeDisplay (shows AI-generated suggestions). The CallContext provides real-time state to these components, including transcript updates, nudges, and call status. The CallDashboard component displays post-call analytics by loading session data.

Key Flow: User interactions → CallContext manages state → WebRTC/WebSocket communication → UI components update in real-time.



Backend Routes

Express.js API route handlers that form the backend's presentation layer. Each route class depends on StateManager for accessing Redis-based call state.

CallRoutes handles the complete call lifecycle (start, end,
session retrieval). TranscriptRoutes manages conversation turns as they're appended during calls. AudioRoutes processes audio chunks through Whisper

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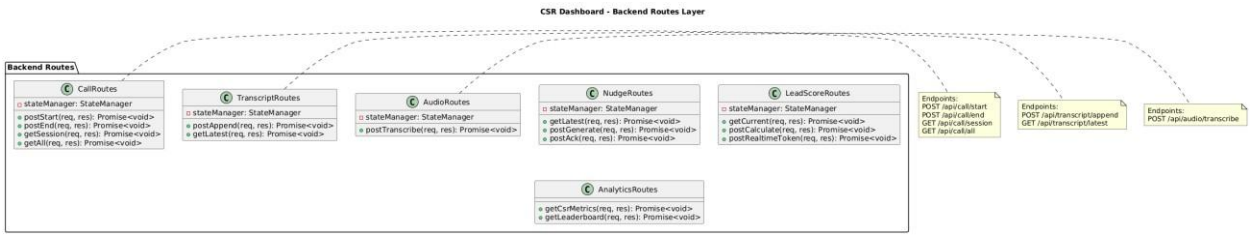
for transcription. NudgeRoutes handles AI-generated suggestion retrieval and acknowledgment. LeadScoreRoutes manages lead scoring calculations and OpenAI Realtime token generation. AnalyticsRoutes provides CSR performance metrics and leaderboards.

Key Flow: HTTP requests → Route handlers → StateManager → Redis/Database → Response back to client.



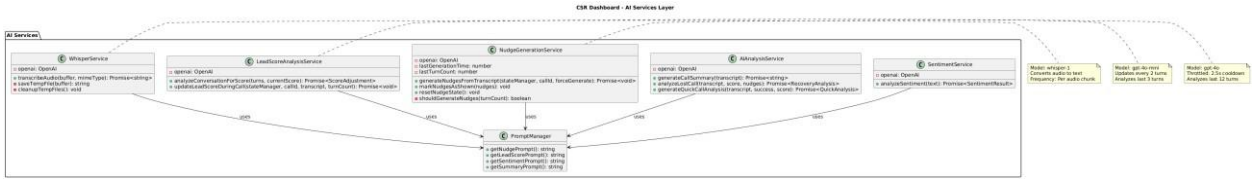
AI_Services

AI-powered services that provide intelligence to the application. All services depend on PromptManager for consistent prompt templates.

WhisperService converts audio chunks to text using OpenAI's Whisper-1 model. AIAnalysisService generates call summaries and recovery strategies using GPT4o. SentimentService analyzes emotional tone of conversations using GPT-4o-mini.

LeadScoreAnalysisService is the most frequently called service—it analyzes the last 3 conversation turns every 2 transcript turns to adjust lead scores dynamically using GPT-4omini. NudgeGenerationService generates real-time sales suggestions by analyzing the last 12 turns with GPT-4o, throttled to run every 2.5 seconds minimum to avoid excessive API calls.

Key Flow: Transcript/Audio data → AI Service → OpenAI API call → Parsed response → Update state/broadcast to clients.



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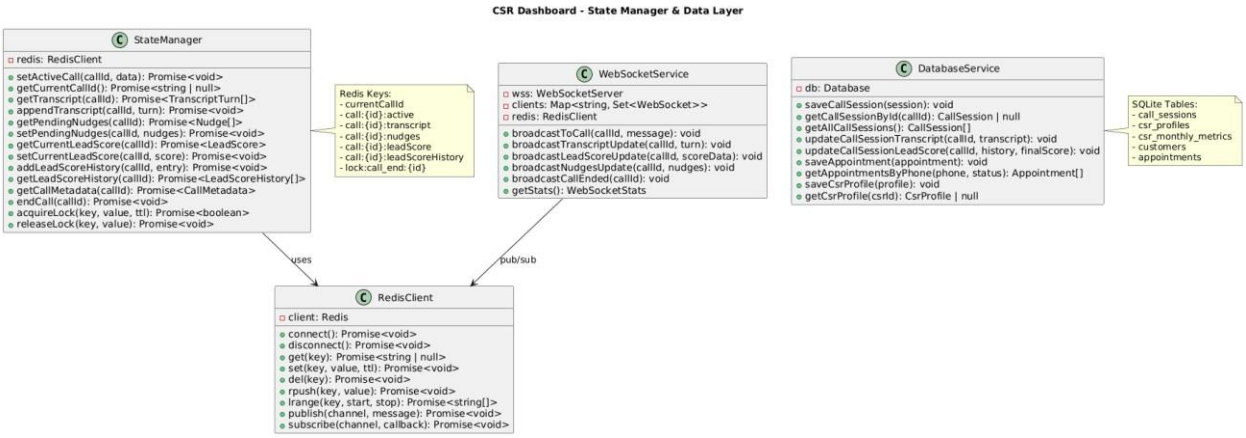
State Manager & Data Layer

Redis stores all active call data with keys like currentCallId, call:{id}:transcript, call:{id}:nudges, and call:{id}:leadScore. This enables fast read/write operations (submillisecond) during active calls. StateManager also implements distributed locking using Redis to prevent race conditions during call end operations.

DatabaseService handles SQLite persistence for historical data, storing complete call sessions, CSR profiles, monthly metrics, customers, and appointments.

WebSocketService uses Redis pub/sub to broadcast real-time updates (transcript changes, lead score updates, nudges) to all connected clients.

Key Flow: Active call data → Redis (hot storage) → Real-time operations. Call ends → Full data saved to SQLite (cold storage) → Redis cleaned up.



Domain Models

This diagram defines the core data structures (domain models) used throughout the application.

CallSession is the central aggregate root containing all call-related data: transcript turns, nudges shown, lead score history, customer information, sentiment analysis, and AIgenerated summaries. It uses composition relationships (filled diamonds) to indicate it owns and manages the lifecycle of

TranscriptTurn,

Nudge, LeadScoreHistory, and CustomerData objects.

TranscriptTurn represents each conversation exchange with role (user/assistant), content, timestamp, and sentiment analysis.

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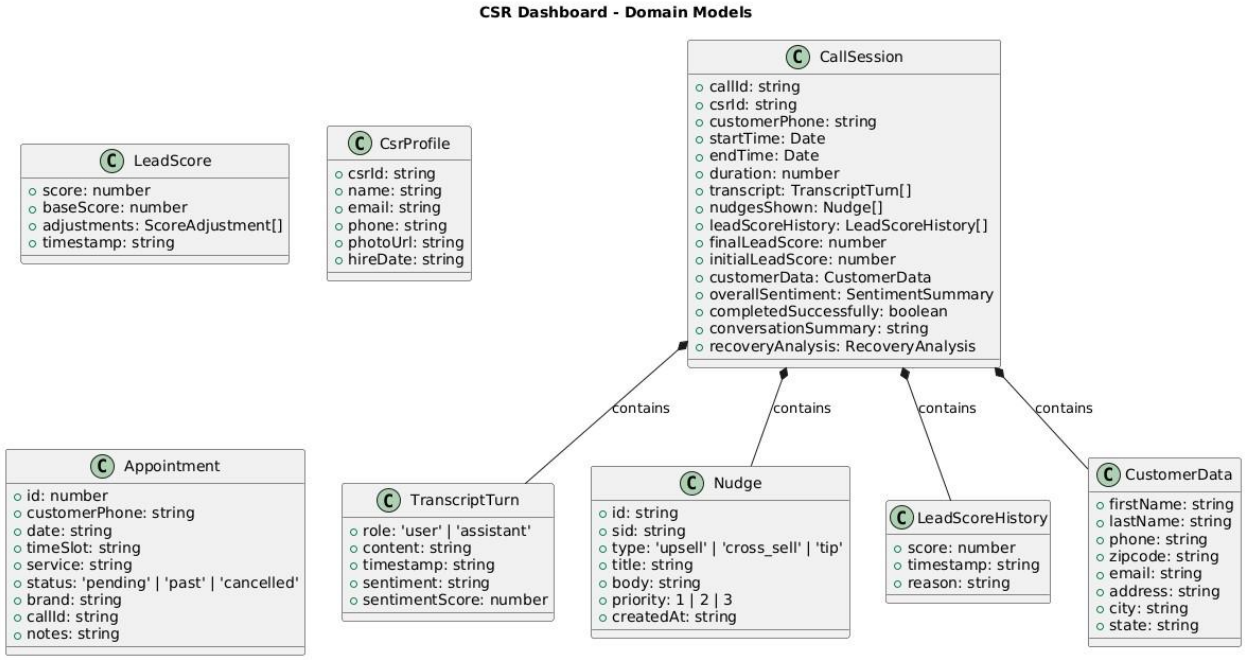
Nudge contains AI-generated sales suggestions with type, priority, and content. LeadScore tracks the current score with base value and all adjustments made during the call. LeadScoreHistory maintains a timeline of score changes for visualization.

CsrProfile and Appointment are independent entities for CSR management and appointment tracking.

Key Flow: Data flows through these models from creation (call start) → updates (during call) → final persistence **(call end) → analytics queries (historical analysis).**